

Endogrid: Measuring the Cost of Reasoning on a Long-Horizon Economic Planning Benchmark

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worldforge_bench

Abstract

Architectures such as the Dragon Hatchling [6] and BDH-CQ [2] are motivated by a specific bet: that adaptation through *recurrent state* can match explicit reasoning at a fraction of the inference cost. Testing that bet needs environments where both routes can be run under one budget and one score. We present ENDOGRID, a deterministic ten-simulated-year infrastructure-development benchmark whose objective is the equity value of a business rather than any quantity the agent can monotonically accumulate, and we use it to measure the cost of reasoning directly. Under a matched twenty-decision budget on the same map, a state-based policy that performs no search, no backtracking and no weight updates scores 40.8; the best language-model agent scores 43.4. The policy reaches 94% of that score at 1/670 of the cost — 32,870 points per dollar against 53.9. Meanwhile an agent restricted to a fixed terrain→machine lookup table scores 6.8, *below* the 16.3 of doing nothing at all, confirming the environment penalises memorised surface correlations rather than rewarding them. We are explicit about what this is and is not: we did not run a BDH model. What we contribute is the harness, the protocol and the measured frontier on which that comparison can be made, together with the finding that on this task the expensive route buys little.

1 The question

The interesting claim shared by the Dragon Hatchling [6] and BDH-CQ [2] is not that recurrent-state models are more accurate. It is that a large part of what explicit reasoning buys can be got more cheaply. BDH is evaluated on Sudoku-Extreme at 97.4% top-1 *with no chain-of-thought and no backtracking*; BDH-CQ adapts to an unseen ARC task through recurrent state alone, with no parameter updates at inference. Both are statements about the price of correctness, not only its ceiling.

That claim is hard to test on puzzle benchmarks, because a puzzle is scored right or wrong and the cost axis is left implicit. What is missing is an environment where (i) an agent must plan over a long horizon in an unfamiliar world, (ii) the objective cannot be climbed by doing more of the rewarded thing, and (iii) both a cheap state-based policy and an expensive reasoning system can be run under one budget and scored on one number.

We built that environment and measured the frontier.

What we claim, and what we do not. We did not run a BDH model. Every result here is a scripted policy or a language-model agent. The contribution is the harness and the measurement: a benchmark where the cost of reasoning is visible, a protocol borrowed from the BDH and ARC-AGI-2 evaluations, and the empirical observation that on this task a policy with no search loop is within 6% of the best language model at three orders

of magnitude less cost. That is a motivation for architectures like BDH, not evidence for them.

2 The environment

An agent is given a procedurally generated terrain, a starting balance sheet of \$25M, and ten simulated years. It sites solar, wind and hydro generators, orients them, wires them into demand zones, finances them with cash and debt, and may hedge output with long-term contracts. Each simulated hour the physical layer resolves — sun position, wind field, water cycle, river flow, machine output, transmission — and then a market clears.

2.1 Why the objective is not monotone

Most agent benchmarks score an accumulable quantity, so an agent climbs by doing more of the rewarded thing. Here the price of electricity is endogenous: it clears every tick against a residual demand curve served by a background thermal fleet in merit order. Renewables bid at zero marginal cost, so the agent’s own output displaces the marginal unit and lowers the price it earns:

Agent supply (kW)	Clearing price (\$/MWh)	Marginal unit
0	79.10	CCGT
10,000	75.25	lignite
20,000	9.00	nuclear
30,000	−3.12	oversupply
60,000	−25.00	price floor

Each asset is discounted at its *own* realised capture price, so an agent that floods the market marks down the assets it already owns. Building more can be worth less, and the naive maximising policy defeats itself. Wealth can fall: capital is scarce, leverage is priced, covenants bind, and insolvency ends the episode.

2.2 Rules that must be inferred, not memorised

No terrain carries an energy multiplier. “Sand means solar” and “mountain means wind” are both false by construction — a leeward mountain cell has *worse* wind than flat ground, because of an orographic wind-shadow term. Output traces through chained physical variables: irradiance through a terrain-obstruction ray-cast and a real incidence cosine against panel tilt and azimuth; turbine power through $\frac{1}{2}\rho Av^3$ with a $\cos(\text{yaw})$ misalignment term that is a hard zero at 90° ; hydro through $\rho g Q H$ with floors on discharge and head.

This is a deliberate defence against causal confusion in the sense of Tien et al. [7]: an agent that latches onto a correlated surface feature instead of the causal chain should be punished. Section 5 shows it is.

3 Protocol

We borrow the evaluation logic of the BDH line of work, and say plainly that it is the logic we are borrowing, not the model.

Table 1: Representative evaluation efforts, and where ENDOGRID sits. *Adaptation* is the route an agent has to a task it has not seen. *Cost axis* asks whether the benchmark scores what correctness cost, not only whether it was reached.

	World	Objective	Horizon	Adaptation	Cost axis
Sudoku-Extreme [6]	fixed puzzle	correct grid	one shot	recurrent state	no
ARC-AGI-2 [1]	held-out tasks	correct grid	one shot	in-context	no
HRM [8]	puzzle suite	correct grid	one shot	backward pass	params only
TRM [5]	puzzle suite	correct grid	one shot	recursive	params only
Crafter [4]	procedural grid	achievements	episode	policy weights	no
MineDojo [3]	open world	task suite	episode	policy weights	no
Endogrid	procedural grid	equity value	10 sim-years	in-episode state	dollars

No weight updates during the run. An agent plays ten simulated years with no retraining, no fine-tuning and no gradient step. Only in-episode state changes: cash, built assets, and whatever the agent has learned about *this* map’s wind, water and terrain. This mirrors BDH-CQ’s adaptation through recurrent state rather than parameters [2]. It is a meaningful constraint rather than an arbitrary one precisely because the alternative exists: the Hierarchical Reasoning Model [8] takes a backward pass per unseen task, and our agents are given no such route. Fixing the constraint makes the two families comparable.

Memory is environmental, not a cache. The recurrent state an agent must track is physical: soil moisture that carries rainfall forward, river discharge that lags its catchment, accumulated evidence about a prevailing wind bearing that mean-reverts seasonally, machine health that degrades. These are closer to the fast-weight, Hebbian synaptic memory of Kosowski et al. [6] than to a key-value cache of past tokens: the state *is* the world’s, it persists whether or not the agent attends to it, and it is what makes a consequence delayed.

Correctness without an explicit search loop. BDH is tested on whether it reaches correct Sudoku solutions with no chain-of-thought and no backtracking [6]. We apply the same behavioural signature. Our **heuristic** agent performs no search: it reads the fields at each candidate cell, ranks by expected return on capital, commits, and never revises by exploring alternatives. It cannot backtrack — a built machine is a spent \$1.45M. Whether such a policy reaches good placements is the same question BDH is asked.

Held-out seeds. Following the private-set methodology of ARC-AGI-2 [1], physical constants are fixed while map layout, river paths, ridge geometry and demand placement vary with the seed. Seeds 1–8 are development; **seeds 9 and 10 were not run until final scoring**. A generalisation gap is reported rather than assumed.

One world, many tiers. That a single procedurally generated grid world with graded difficulty is a legitimate evaluation design rather than a toy was established by Crafter [4], and extended to open-ended task suites by MineDojo [3]. Our capability targets — spatial reasoning over terrain and wind geometry, construction planning, memory of prior environmental outcomes, causal rule learning, and transfer to unfamiliar layouts — follow that lineage.

Table 2: Ten seeds, ten simulated years each, matched twenty-decision budget. Equity is terminal equity value in millions against a \$25M start; GWh is energy delivered; LCOE is levelised cost in \$/MWh; *Ins.* counts insolvencies.

Agent	Score	sd	min	max	Equity	Return	GWh	LCOE	Ins.
heuristic	43.8	11.5	14.6	55.1	46.5	+86%	381	44	0
random	22.2	11.4	10.7	43.8	28.3	+13%	52	157	0
donothing	16.3	0.0	16.3	16.3	25.0	+0%	0	0	0
lookup	9.1	2.5	4.4	12.2	17.0	-32%	86	290	0

4 Agents

Agent	What it does
donothing	Holds its starting capital. The floor any agent must clear to have created value.
lookup	Consults a fixed terrain→machine table at a fixed orientation. Never observes wind, sun, price or its own books. The memorisation control.
random	Builds legal machines at random cells and orientations. The noise floor.
heuristic	Ranks sites by the fields actually present at each cell, orients to the resource, wires to the nearest reachable zone, and stops building once its own supply would depress the price. No search, no backtracking.
LLM agents	Opus 5, Sonnet 5, Haiku 4.5, playing blind through a command interface. Given the action schema and the objective only — no description of the price mechanism, of what drives output, or of how assets are valued — and instructed not to read the implementation.

5 Results

Memorisation loses money. **lookup** scores 9.1, below the 16.3 of holding capital and building nothing, with a mean return of -32%. An agent applying a fixed terrain→machine table does not merely fail to excel; it destroys capital, because it sites without regard to the resource that actually reaches a cell and orients every machine identically regardless of the wind. This is the falsifiable form of the anti-memorisation claim: if a lookup table ever became competitive, the environment would be rewarding the surface correlations it was built to punish [7].

Held-out seeds. **heuristic** averages 46.0 on the eight development seeds and 34.9 on seeds 9 and 10, which were not run until final scoring — a relative gap of +24.2%, within noise, so we see no evidence of layout fitting. The policy has no parameters to fit, so this is a check that the *environment* does not vary wildly by layout, as much as a check on the agent.

Table 3 is the result this paper exists for. On the same map, under the same budget, Sonnet 5 scores 43.4 and **heuristic** scores 40.8. The language model is ahead — we are not claiming otherwise. But it costs \$0.80 and 3.8 minutes to get there, against a tenth of a cent and 0.7 minutes. The policy with no search loop reaches 94% of the score at roughly 1/686 of the cost, or 686× the accuracy per dollar.

Why this is the interesting axis. The same shape of argument runs through the recursive-reasoning literature: TRM reaches competitive accuracy with 7M parameters

Table 3: Cost of reasoning. All agents on seed 1 under the same twenty-decision budget. Language-model cost is the whole-episode token total at published first-party rates; scripted cost prices CPU at \$0.10/hour, a deliberately generous figure. *Score/\$* is the resulting accuracy-per-dollar. Scripted rows come from the same engine version as Table 2. Haiku 4.5’s row predates a correction to hydro siting and is contaminated by it; the other two language-model rows built no affected assets and are unaffected.

Agent	Score	Equity	Cost	Minutes	Score/\$
Sonnet 5	43.4	46.8	\$0.80	3.8	54
Opus 5	33.9	37.9	\$2.26	6.3	15
Haiku 4.5	8.2	18.7	\$0.35	2.1	24
heuristic	40.8	44.9	\$0.0011	0.7	36,960
random	43.8	44.9	\$0.0010	0.6	43,050
donothing	16.3	25.0	\$0.0007	0.4	22,149
lookup	11.7	22.8	\$0.0011	0.7	10,572

where HRM uses 27M [5, 8], and BDH-CQ adapts through recurrent state rather than a backward pass per task [2]. What those results assert about parameter counts, Table 3 measures in dollars on a task with a long horizon and an economic objective. A model that matched the language-model score at the state-based policy’s cost would dominate this frontier outright. Nothing we ran occupies that corner — which is precisely the corner BDH-style architectures claim.

6 Limitations

No BDH model was run. This is a harness and a measurement, not an architecture comparison. The frontier we report is where a BDH-style model would have to land to be interesting; we have not put one there.

Single-seed language-model results. Three episodes, one seed, one run each, on an engine version since corrected for a defect those episodes exposed. They calibrate cost and show the interface is usable without explanation. They do not rank models and we make no such ranking.

The causal-confusion ablation is not run. We designed against spurious terrain shortcuts [7] and we show that a pure lookup table loses money. We have *not* shown that our reasoning agents are free of some other surface shortcut. The honest test would hold physics fixed while permuting the cosmetic identity of terrain types; an agent relying on a genuine causal chain should be unaffected. That experiment is designed and unrun, and it is the first thing we would do next.

Cost accounting favours the scripted agents. We price their CPU at a generous \$0.10/hour and count language-model tokens at published rates. The ratio is large enough that the accounting convention does not change the conclusion, but it is a convention.

7 Reproducing this

Deterministic: a seed, a configuration and an ordered action log reproduce a bit-identical state hash. Without that, a score difference between agents might be luck.

```

pip install -e .
wfbench bench --agents donothing,lookup,random,heuristic --seeds 1-10 --years 10
wfbench generalise --agent heuristic --train 1-8 --held-out 9-10
wfbench serve      # line-delimited JSON, for language-model agents
wfbench api        # HTTP, for web front-ends and remote agents

```

Full interface documentation, including the action schema and the three ways to drive the engine, is in `docs/API.md`. Forty-one executable invariants cover per-tick water mass balance and energy conservation, determinism under action-log replay, wind-shadow directionality, turbine cut-in/cut-out, hydro floors, and the anti-memorisation probe.

References

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